THE UNIVERSITY of EDINBURGH
School of Biological Sciences

COVER SHEET

Comparative and Evolutionary Genomics: Using Phylogenetic Profiles to Predict Functional Linkage

In-Course Assessment 2

Student Exam Number:	B269948

Declaration	
I **have/have not** used Generative AI-based methods (e.g. ChatGPT) in the preparation of this Dissertation.	**I have**

I used ChatGPT to understand the difficult concepts further and debug the scripts I have written.

Word Count*	1821

*The word count should include figure and table legends but not include the tables themselves and not include references.

Methods

A. Organism selection and obtaining genome wide protein sets.

1. I selected 10 taxa from Cyanobacteria species, 5 of them were reported to have the ability to fix nitrogen and 5 of them were not in the paper written by Watanabe and Horiike (1).
2. I searched the taxa on NCBI Genome database (2). I checked if the data was contaminated. If not, I selected the data and clicked “Download” and chose “Download Package”. After that, I chose “RefSeq only” and “Protein (FASTA)” options and downloaded the dataset.
3. The dataset was downloaded in a zip file. I found the fasta file regarding the protein set and renamed it according to the organism to make the source clear to myself. I recommend doing this to avoid extra confusion.

I am presenting the information regarding the taxa I chose, their ability to fix nitrogen and the accession number on Genome database on Table 1 below. I chose them according to the information from the paper of Watanabe and Horiike (1).

Table 1: Information regarding the name of the taxa, their ability to fix nitrogen and accession numbers on Genome database (2) according to the paper (1).

Name of the taxa	Ability to fix nitrogen	Accession number on Genome database
Halothece sp. PCC 7418	Yes	GCF_000317635.1
Euhalothece natronophila Z-M001	Yes	GCF_007904085.1
Leptolyngbya sp. O-77	Yes	GCF_001548395.1
Trichodesmium erythraeum IMS101	Yes	GCF_000014265.1
Synechococcus sp. JA-3-3Ab	Yes	GCF_000013205.1
Thermosynechococcus sp. CL-1	No	GCF_008386235.1
Limnospira fusiformis SAG 85.79	No	GCF_012516315.1
Geitlerinema sp. PCC 7407	No	GCF_000317045.1
Synechococcus elongatus PCC 11801	No	GCF_003846445.2
Cyanobium sp. NIES-981 strain NIES-981	No	GCF_900088535.1

B. Orthogroup predictions.

1. I ran OrthoFinder (version 2.5.4) (3) software on the datasets to determine the orthogroups. I set the parameters of threads to 2, inflation parameter to 1.4. I included the option of “og” to stop the software after finding the orthogroups. I used lecture material (4) as guidance.

C. Creating phylogenetic profiles from protein sets.

1. I used the code that was given to us in lecture material of Comparative and Evolutionary Genomics by the lecturer Daniel Barker (makephyprofiles.sh) (5) to obtain phylogenetic profiles of the protein sets. I added a line to the code to print the results into a file.
2. I copied the code file into the results file of OrthoFinder (3) to make it easier, the input file of the software was “Orthogroups.GeneCount.tsv” file which is an output of OrthoFinder (3) software.

D. Obtaining the profile for *nifH* gene

1. I created a file with all of the protein sets. Then I performed a keyword search with "nitrogenase iron protein". This returned 5 proteins.
2. I took the sequence of the protein coding for the entry "WP_068515290.1 MULTISPECIES: nitrogenase iron protein [*Cyanophyceae]*" from the Genome database data (2) and searched on the InterPro Database (6).
3. I ran a BLASTP (version 2.13.0)(7) search on the sequence against the file with all of the protein sequences that I created in step 1.
4. I looked up the orthogroups for each of the top 10 hits.
5. I keyword searched the name of the orthogroup containing *nifH* in profiles file I created in Section C. I took the profile of the orthogroup and saved it into another file.

E. Obtaining the orthogroups that have profiles similar to *nifH* profile.

1. To obtain profiles similar to nifH, I wrote a script on RStudio (version 2022.2.1.461-Prairie Trillium) (8) (R version 4.1.3)(9). I calculated Hamming distance of orthogroup profiles to the nifH orthogroup profile using lecture material (10) as guidance.
2. I selected the orthogroups with Hamming score of 0, and using my script, I wrote them to files.
3. I keyword searched the orthogroups file to determine the protein sequences in those orthogroups.
4. I ran one of the sequences of each group on InterPro database (6) to gain information regarding their family.
5. I classified the orthogroups based on their functions.

Results

The results I obtained from the experiment I carried out which I described in Methods section are divided according to the sections.

B. Orthogroup predictions

I ran the OrthoFinder (3) software on 10 genome wide protein sets. The software reported the total number of genes to be 36971. Of those, it was able to assign 32142 of them to orthogroups (86.9%)(3). It created 4692 orthogroups, 487 of the orthogroups were reported to be species-specific (4.9%)(3).

D. Obtaining the profile for *nifH* gene

I performed the keyword search as I described in the Methods part (Section D). I chose the protein "WP_068515290.1 MULTISPECIES: nitrogenase iron protein [*Cyanophyceae]*" to look for the *nifH* profile. The orthogroup it was in was called "OG0002638". The table below shows the result regarding this orthogroup.

Table 2: Protein accession numbers found in the orthogroup OG0002638 according to the organism's name, taken from the "Orthogroups.csv" file from output of OrthoFinder (3) software. The names of the proteins were taken from the combined protein file, created from data of Genome database (2).

Organism Name	**Protein Accession Number (if it is found in the Orthogroup OG0002638)**	**Protein Name (if it is found in the Orthogroup OG0002638)**
Halothece sp. PCC 7418	WP_015225829.1	nitrogenase iron protein [*Halothece* sp. PCC 7418]
Euhalothece natronophila Z-M001	WP_146294449.1	nitrogenase iron protein [*Euhalothece natronophila]*
Leptolyngbya sp. O-77	WP_068515290.1	MULTISPECIES: nitrogenase iron protein [*Cyanophyceae]*
Trichodesmium erythraeum IMS101	WP_011613474.1	nitrogenase iron protein [*Trichodesmium erythraeum*]
Synechococcus sp. JA-3-3Ab	WP_011430651.1	MULTISPECIES: nitrogenase iron protein [unclassified *Synechococcus*]
Thermosynechococcus sp. CL-1	-	-
Limnospira fusiformis SAG 85.79	-	-
Geitlerinema sp. PCC 7407	-	-
Synechococcus elongatus PCC 11801	-	-
Cyanobium sp. NIES-981 strain NIES-981	-	-

To ensure that this is the *nifH* gene, I ran the sequence of WP_068515290.1 protein on InterPro database (6). The result I obtained stated the protein belonged to NifH/frxC family and its Nitrogenase iron protein NifH family (6).

Further, I ran BLASTP (7) on the combined dataset of proteins using the sequence of WP_068515290.1 as query. The results I obtained are in Table 3 below (showing only top 5 hits). I keyword searched the proteins to find their orthogroups. The top 10 hits are included in supplementary data (Table S1).

Table 3: Top 5 hits of WP_068515290.1 protein sequence (2) from BLASTP (7).

Protein Name	Bit Score	e-value	Orthogroup
WP_068515290.1 MULTISPECIES: nitrogenase iron protein [*Cyanophyceae*]	595	0.0	OG0002638
WP_015225829.1 nitrogenase iron protein [*Halothece* sp. PCC 7418]	504	0.0	OG0002638
WP_011430651.1 MULTISPECIES: nitrogenase iron protein [unclassified *Synechococcus*]	483	1e-174	OG0002638
WP_011613474.1 nitrogenase iron protein [*Trichodesmium erythraeum*]	462	3e-166	OG0002638
WP_146294449.1 nitrogenase iron protein [*Euhalothece natronophila*]	461	6e-166	OG0002638

I ensured the protein I decided to use (WP_068515290.1) (2) was *nifH* gene therefore, I took the phylogenetic profile for the orthogroup containing the protein. The profile is given in the table below.

Table 4: Phylogenetic profile for orthogroup OG0002638.

Species (Name of the taxa)	OG0002638
Halothece sp. PCC 7418	1
Euhalothece natronophila Z-M001	1
Leptolyngbya sp. O-77	1
Trichodesmium erythraeum IMS101	1
Synechococcus sp. JA-3-3Ab	1
Thermosynechococcus sp. CL-1	0
Limnospira fusiformis SAG 85.79	0
Geitlerinema sp. PCC 7407	0
Synechococcus elongatus PCC 11801	0
Cyanobium sp. NIES-981 strain NIES-981	0

E. Obtaining the orthogroups that have profiles similar to *nifH* profile.

After running my script, I was able to obtain 14 orthogroups (apart from the *nifH* itself) that had Hamming distance value of 0 according to OG0002638. I selected a protein to represent the orthogroup and searched them on the InterPro database (6) to understand their functions and families. The results for first 10 orthologroups I obtained are in the table below. The rest of the orthologroups are included in supplementary data (Table S2).

Table 5: The first 10 orthogroups that have 0 Hamming Distance, the protein accession number to be chosen as representative and the family of the protein according to InterPro Database (6).

Orthogroup	**Representative Protein**	**Protein Name**	**Protein Family**
OG0002633	WP_146294435.1	helix-turn-helix domain-containing protein [*Euhalothece natronophila]*	None predicted
OG0002634	WP_015225820.1	putative nitrogen fixation protein NifT [*Halothece* sp. PCC 7418]	NifT/FixU
OG0002636	WP_011613471.1	nitrogenase cofactor biosynthesis protein NifB [*Trichodesmium erythraeum*]	Nitrogenase cofactor biosynthesis protein NifB
OG0002637	WP_011430652.1	MULTISPECIES: Fe-S cluster assembly protein NifU [unclassified *Synechococcus*]	NIF system FeS cluster assembly, NifU
OG0002639	WP_068515287.1	nitrogenase molybdenum-iron protein alpha chain [*Leptolyngbya* sp. O-77]	Nitrogenase molybdenum-iron protein alpha chain
OG0002640	WP_146294453.1	nitrogen fixation protein NifZ [*Euhalothece natronophila*]	Nitrogenase MoFe maturation protein NifZ
OG0002641	WP_015225833.1	nitrogenase iron-molybdenum cofactor biosynthesis protein NifE [*Halothece* sp. PCC 7418]	Nitrogenase MoFe cofactor biosynthesis protein NifE
OG0002642	WP_015225835.1	nitrogen fixation protein NifX [*Halothece* sp. PCC 7418]	Nitrogen fixation protein NifX
OG0002643	WP_068515272.1	NifX-associated nitrogen fixation protein [*Leptolyngbya* sp. O-77]	NifX-associatted nitrogen fixation protein
OG0002644	WP_011430666.1	nitrogenase-stabilizing/protective protein NifW [unclassified *Synechococcus*]	Nitrogen fixation protein NifW

The profiles of the sequences that I obtained from the BLASTP (7) that were not in top 5. Therefore, I keyword searched the orthogroups the proteins belonged to which was OG0000924. I calculated the Hamming distance of this orthogroup to the *NifH* orthogroup using the script I wrote, and the distance was 5.

I included the full list for each orthogroup in supplementary table 3 (Table S3). Since the representative protein I chose for orthogroup OG0002633 were found not to be in a predicted protein family, I decided to inspect the other proteins on the InterPro database (6) as well. None of the other proteins returned results for protein family. All were reported to have a 4Fe-4S ferredoxin-type iron-sulfur binding domain profile (6).

Discussion

The phylogenetic profiling is a useful method to predict proteins functional linkages or characterisation of proteins (11). It is also an important tool to understand the evolutionary preservation of such proteins (11). In this paper, this method was used to examine the NifH proteins and to identify potential proteins that are associated with the nitrogen fixation pathway in accordance with the data on cyanobacteria from the article by Watanabe and Horiike (1).

To summarise the results, I examined 10 different taxa of cyanobacteria; 5 of them were known to be able to fix nitrogen and 5 of them were not (1). I selected the orthogroup (OG0002638) containing nitrogenase iron proteins to be predicted *NifH* orthogroup and ran analyses to ensure they were correctly annotated. Further analysis I have performed via InterPro database (6) on the protein I chose (WP_068515290.1) have confirmed the protein family which was reported to be nitrogenase iron protein NifH family. Moreover, performing BLASTP (7) analysis on the protein I selected confirmed that the other proteins in the orthogroup were closely related to the query sequence (Table 3) and there is a significant difference in bit scores between top 5 (most being 134) and the rest of the top 10 (least being 435) hits list (Table 3 and S1). Using the phylogenetic profile of this orthogroup (Table 4), I was able to find 14 orthogroups that had identical phylogenetic profile (Hamming score of 0)(10) excluding the predicted *NifH* orthogroup (Table 5 and Table S2). I chose one protein to represent the orthogroup and performed a sequence-based search on InterPro database (6) and reported the information regarding their protein families (Table 5 and Table S2).

First, the orthogroup (OG0002638) included 5 proteins which were from each species that were known to fix nitrogen (Table 1 and 2) (1). This was expected as the species without the ability to fix nitrogen are found to have lost the genes regarding nitrogen fixation (1).

Second, the main role players in nitrogenase fixation pathway in cyanobacteria are known. One of the most important molecules in fixation are Mo-type nitrogenase enzymes which contains the subunit of MoFe protein (12). The proteins in the orthogroups OG0002640 and OG0002641 are directly related to MoFe protein. Moreover, as stated in the paper (12), proteins NifW, NifZ and NifX carry importance for the pathway especially in cyanobacteria. I was able to detect 4 orthogroups related to these proteins; OG0002642 and OG0002643 (NifX), OG0002644 (NifW) and OG0002640 (NifZ). Additionally, I was able to locate the orthogroups containing proteins associated to NifB (WP_011613471.1) and NifU (OG0002637) which are important for FeMo-co complex (12). FeMo-co stands for iron-molybdenum cofactor and is a part of MoFe protein (12). This complex is important for nitrogen fixation biosynthesis (12). Further, OG0002639 and OG0002680 orthogroups are reported to be related to iron-molybdenum protein, and upon the InterPro (6) search on OG0002633 representative protein, I was able to say that it carried 4Fe-4S ferredoxin-type iron-sulfur binding domain profile which connected the orthogroup to nitrogen fixation pathway (12). Indirectly, the orthogroup OG0002646 is related to the pathway because of transportation of important nitrogen sources: nitrate and nitrite which are transported via ABC transporters (13).

Interestingly, the orthogroups that are not directly connected to the pathway were OG0002675 and OG0002679. These contained proteins related to AtzE and AtzG proteins. It was reported in this paper that the formation of active state of AtzE protein required AtzG protein (14). AtzE protein is important for synthesis of nitrogen from atrazine which is a type of herbicide (15). As researched in this paper (16), the cyanobacteria with the ability to fix nitrogen could be used for eliminating residues of herbicides from the soil and the crops.

In conclusion, I would say the phylogenetic profiling analysis was successful in predicting functional linkage considering out of 14 orthogroups that were found to have identical profile to *nifH* gene, only 3 were indirectly connected to the pathway. For future work, I would recommend performing an analysis with a maximum likelihood model which could lead to more accurate predictions as suggested by Barker and Pagel (17).

References

1. Watanabe T, Horiike T. The evolution of molybdenum dependent nitrogenase in cyanobacteria. Biology (Basel). 2021;10(4).
2. Genome [Internet]. Bethesda (MD): National Library of Medicine (US), National Center for Biotechnology Information; 2004 – [cited 2025 04 21]. Available from: https://www.ncbi.nlm.nih.gov/datasets/genome/.
3. Emms DM, Kelly S. OrthoFinder: Phylogenetic orthology inference for comparative genomics. Genome Biol. 2019;20(1).
4. Barker D. Week 2 Tutorial: Delimiting and assessing orthogroups [unpublished lecture notes]. PGBI11115: Comparative and Evolutionary Genomics, University of Edinburgh; lecture given 2025, January 22.
5. Barker D (2025) makephyprofiles.sh. Unpublished coursework material, Comparative and Evolutionary Genomics (PGBI11115), University of Edinburgh.
6. Blum M, Andreeva A, Florentino LC, Chuguransky SR, Grego T, Hobbs E, et al. InterPro: the protein sequence classification resource in 2025. Nucleic Acids Res [Internet]. 2025 Jan 6;53(D1):D444–56. Available from: http://www.ncbi.nlm.nih.gov/pubmed/39565202
7. Altschul SF, Madden TL, Schäffer AA, Zhang J, Zhang Z, Miller W, et al. Gapped BLAST and PSI-BLAST: A new generation of protein database search programs. Vol. 25, Nucleic Acids Research. 1997.
8. RStudio Team. RStudio: Integrated Development Environment for R. Boston, MA: RStudio, PBC; 2022.
9. R Core Team (2022). R: A Language and Environment for Statistical Computing_. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
10. Barker D. Week 5 Tutorial: Phylogenetic Profiles [unpublished lecture notes]. PGBI11115: Comparative and Evolutionary Genomics, University of Edinburgh; lecture given 2025, February 12.
11. Pellegrini M, Marcotte EM, Thompson MJ, Eisenberg D, Yeates TO. Assigning protein functions by comparative genome analysis: Protein phylogenetic profiles. Proc Natl Acad Sci U S A. 1999;96(8).
12. Fujita Y, Uesaka K. Nitrogen fixation in cyanobacteria. In: Cyanobacterial Physiology: From Fundamentals to Biotechnology. 2022.
13. Seth K, Kumawat G, Kumar M, Sangela V, Singh N, Gupta AK, et al. Nitrogen Metabolism in Cyanobacteria. In: Ecophysiology and Biochemistry of Cyanobacteria. 2022.
14. Esquirol L, Peat TS, Wilding M, Liu JW, French NG, Hartley CJ, et al. An unexpected vestigial protein complex reveals the evolutionary origins of an s-triazine catabolic enzyme. Journal of Biological Chemistry. 2018;293(20).
15. Govantes F, Porrúa O, García-González V, Santero E. Atrazine biodegradation in the lab and in the field: Enzymatic activities and gene regulation. Vol. 2, Microbial Biotechnology. 2009.
16. Yadav S, Srivastava R, Singh N, Kanda T, Verma E, Choudhary P, et al. Cyanobacteria-Pesticide Interactions and Their Implications for Sustainable Rice Agroecosystems. Callaway TR, editor. Int J Microbiol [Internet]. 2025 Jan;2025(1). Available from: https://onlinelibrary.wiley.com/doi/10.1155/ijm/7265036
17. Barker D, Pagel M. Predicting functional gene links from phylogenetic-statistical analyses of whole genomes. PLoS Comput Biol. 2005;1(1).

Supplementary Data

Table S1: Top 10 hits of WP_068515290.1 (2) protein sequence from BLASTP (7).

Protein Name	Bit Score	e-value	Orthogroup
WP_068515290.1 MULTISPECIES: nitrogenase iron protein [*Cyanophyceae*]	595	0.0	OG0002638
WP_015225829.1 nitrogenase iron protein [*Halothece* sp. PCC 7418]	504	0.0	OG0002638
WP_011430651.1 MULTISPECIES: nitrogenase iron protein [unclassified *Synechococcus*]	483	1e-174	OG0002638
WP_011613474.1 nitrogenase iron protein [*Trichodesmium erythraeum*]	462	3e-166	OG0002638
WP_146294449.1 nitrogenase iron protein [*Euhalothece natronophila]*	461	6e-166	OG0002638
WP_087067474.1 ferredoxin:protochlorophyllide reductase (ATP-dependent) iron-sulfur ATP-binding protein [*Cyanobium* sp. NIES-981]	160	1e-47	OG0000924
WP_208673436.1 ferredoxin:protochlorophyllide reductase (ATP-dependent) iron-sulfur ATP-binding protein [*Synechococcus elongatus*]	139	2e-39	OG0000924
WP_149819367.1 MULTISPECIES: ferredoxin:protochlorophyllide reductase (ATP-dependent) iron-sulfur ATP-binding protein [unclassified *Thermosynechococcus*]	139	2e-39	OG0000924
WP_041439195.1 MULTISPECIES: ferredoxin:protochlorophyllide reductase (ATP-dependent) iron-sulfur ATP-binding protein [unclassified *Synechococcus*]	138	4e-39	OG0000924
WP_006624178.1 MULTISPECIES: ferredoxin:protochlorophyllide reductase (ATP-dependent) iron-sulfur ATP-binding protein [*Limnospira*]	134	1e-47	OG0000924

Table S2: The orthogroups that have 0 Hamming Distance, the protein accession number to be chosen as representative and the family of the protein according to InterPro Database (6).

Orthogroup	**Representative Protein**	**Protein Name**	**Protein Family**
OG0002633	WP_146294435.1	helix-turn-helix domain-containing protein [*Euhalothece natronophila]*	None predicted
OG0002634	WP_015225820.1	putative nitrogen fixation protein NifT [*Halothece* sp. PCC 7418]	NifT/FixU
OG0002636	WP_011613471.1	nitrogenase cofactor biosynthesis protein NifB [*Trichodesmium erythraeum*]	Nitrogenase cofactor biosynthesis protein NifB
OG0002637	WP_011430652.1	MULTISPECIES: Fe-S cluster assembly protein NifU [unclassified *Synechococcus*]	NIF system FeS cluster assembly, NifU
OG0002639	WP_068515287.1	nitrogenase molybdenum-iron protein alpha chain [*Leptolyngbya* sp. O-77]	Nitrogenase molybdenum-iron protein alpha chain
OG0002640	WP_146294453.1	nitrogen fixation protein NifZ [*Euhalothece natronophila*]	Nitrogenase MoFe maturation protein NifZ
OG0002641	WP_015225833.1	nitrogenase iron-molybdenum cofactor biosynthesis protein NifE [*Halothece* sp. PCC 7418]	Nitrogenase MoFe cofactor biosynthesis protein NifE
OG0002642	WP_015225835.1	nitrogen fixation protein NifX [*Halothece* sp. PCC 7418]	Nitrogen fixation protein NifX
OG0002643	WP_068515272.1	NifX-associated nitrogen fixation protein [*Leptolyngbya* sp. O-77]	NifX-associatted nitrogen fixation protein
OG0002644	WP_011430666.1	nitrogenase-stabilizing/protective protein NifW [unclassified *Synechococcus*]	Nitrogen fixation protein NifW

OG0002646	WP_011613447.1	glycine betaine/L-proline ABC transporter ATP-binding protein [*Trichodesmium erythraeum*]	ABC transporter osmolyte uptake ATP-binding protein, Glycine betaine/proline betaine transport system ATP-binding protein ProV-like
OG0002675	WP_068508804.1	Asp-tRNA(Asn)/Glu-tRNA(Gln) amidotransferase GatCAB subunit A [*Leptolyngbya* sp. O-77]	1-carboxybiuret hydrolase, AtzE subunit
OG0002679	WP_011430915.1	DUF4089 domain-containing protein [unclassified *Synechococcus*]	1-carboxybiuret hydrolase subunit AtzG-like
OG0002680	WP_011613476.1	nitrogenase molybdenum-iron protein subunit beta [*Trichodesmium erythraeum*]	ChlB/BchB/BchZ Nitrogenase molybdenum-iron protein beta chain

Table S3: The protein accession numbers for the orthogroups provided by OrthoFinder (3).

Orthogroups	**Protein Accession IDs**
OG0002633	WP_146294435.1, WP_015225297.1, WP_068515247.1, WP_011611516.1
OG0002634	WP_146294440.1, WP_015225820.1, WP_068515251.1, WP_011430645.1, WP_011613466.1
OG0002636	WP_146294445.1, WP_015225825.1, WP_068515299.1, WP_011430655.1, WP_011613471.1
OG0002637	WP_146294448.1, WP_015225828.1, WP_068515292.1, WP_011430652.1, WP_011613473.1
OG0002639	WP_146294450.1, WP_015225830.1, WP_068515287.1, WP_011430650.1, WP_011613475.1
OG0002640	WP_146294453.1, WP_015225832.1, WP_068515283.1, WP_011430646.1, WP_011613467.1
OG0002641	WP_146294454.1, WP_015225833.1, WP_084782916.1, WP_011430661.1, WP_011613477.1
OG0002642	WP_146294456.1, WP_015225835.1, WP_068515278.1, WP_011430663.1, WP_011613478.1
OG0002643	WP_146294457.1, WP_015225836.1, WP_068515272.1, WP_011430664.1, WP_011613479.1
OG0002644	WP_146294458.1, WP_015225838.1, WP_068515304.1, WP_011430666.1, WP_011613480.1
OG0002646	WP_146294686.1, WP_015226247.1, WP_068512523.1, WP_011431042.1, WP_011613447.1
OG0002675	WP_146297188.1, WP_041596117.1, WP_068508804.1, WP_011430914.1, WP_011613535.1
OG0002679	WP_186709000.1, WP_015224561.1, WP_068508802.1, WP_011430915.1, WP_011610368.1
OG0002680	WP_222597884.1, WP_216086731.1, WP_068515285.1, WP_011430649.1, WP_011613476.1
OG0002638 (*NifH* Orthogroup)	WP_068515290.1, WP_015225829.1, WP_011430651.1, WP_011613474.1, WP_146294449.1